5TH INTERNATIONAL CONFERENCE

Smart Grid Synchronized Measurements & *Analytics*

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Benchmarking Dynamic Frequency Estimators for Low-Inertia IBR Grids

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Operating claim

- Frequency estimation under grid disturbances is a **multi-objective measurement problem**.
- Rankings change with the objective: frequency RMSE, RoCoF error, latency, CPU cost, and disturbance sensitivity produce different orders.
- IBR-era disturbances make this operational: distorted waveforms and fast controls can turn a measurement artifact into a ride-through or forensic diagnosis problem.
- Estimator choice is conditional: event family, objective metric, and compute budget determine the preferred method.

Measurement rule

Report error, transient exposure, settling, latency, and cost as separate outputs for each estimator.

Experiment scale

9+2
estimators

4
families

5
scenarios

grid
search

Observed pattern

RA-EKF improves dynamic tracking and false-trip exposure under IBR composite stress; protection-class frequency measurement needs tests beyond compliance-style cases.

Dataset: 9 principal estimators, 2 legacy baselines, 5 stress scenarios, standard and IBR-specific metrics, plus an expanded sensitivity sweep.

Technical contributions

Scope

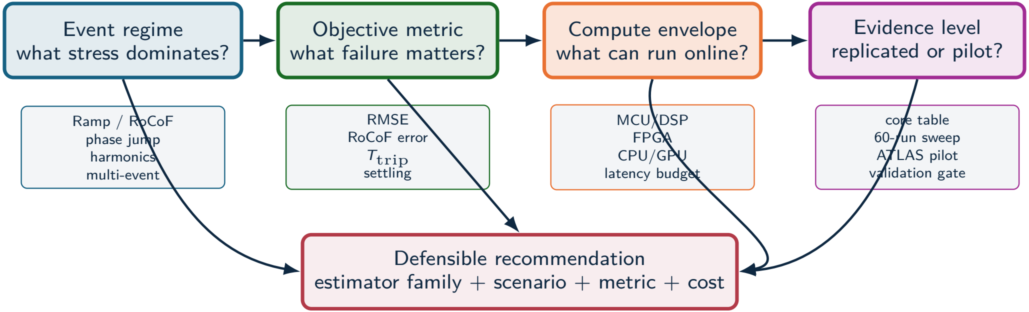
- PMU time-stamp certification;
- a single universal leaderboard;
- hardware latency measurements.

Contributions

- a stress-test benchmark for relay-class frequency estimators;
- an adapted RA-EKF with explicit $\dot{\omega}$ state, covariance scaling, and event gating;
- a metric-complete comparison: RMSE, peak error, T_{trip}, settling, and CPU;
- evidence that standard compliance tests miss IBR failure modes.

The stress-test protocol links RA-EKF design choices to phase-jump false-trip exposure.

Estimator selection logic



Example: RA-EKF is recommended for ramp/phase-jump protection metrics under DSP/FPGA-class cost; global RMSE leadership remains scenario-dependent.

IBR disturbances expose the gap

Event	Observed behavior	Monitoring limitation	Benchmark implication
Blue Cut Fire, CA 2016	Transmission faults caused nearly 1,200 MW of solar PV output loss; NERC linked part of the behavior to near-instantaneous frequency or voltage response.	The critical interval was a distorted, fault-clearing transient on a subsecond timescale.	Test whether estimators confuse waveform distortion with real frequency collapse.
Canyon 2 Fire, CA 2017	Two cleared faults produced 682 MW and 937 MW solar PV reductions.	NERC noted 4 s SCADA could not distinguish momentary cessation from tripping in some cases.	Report latency, settling, invalid samples, event-window behavior, and aggregate error.
Odessa events, TX 2021–2022	More than 1 GW-scale solar reductions followed normally cleared faults far from affected plants.	NERC found controls, protection, and model-fidelity gaps, including effects not captured well by positive-sequence models.	Stress tests must include control-like discontinuities, phase jumps, and model-mismatch regimes.
California BESS, 2022	Normally cleared faults caused abnormal BESS active-power reductions and inverter trips.	Causes included ac overcurrent/overvoltage, dc bus issues, unbalanced currents, and data-quality limits.	Evaluate frequency estimators under unbalance and non-sinusoidal contamination.

Source: NERC disturbance reports; NERC IBR disturbance monitoring white paper

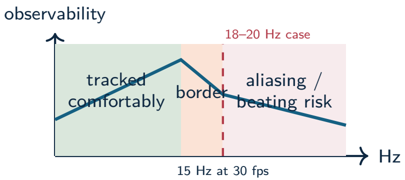
Standard PMU stream limits

Fast fault and ride-through

IBR controls can react within cycles. A low-rate or heavily filtered frequency estimate may miss the subcycle cause while still showing the delayed consequence.

Oscillation near reporting-rate limits

The Kaua'i 2021 event exhibited 18–20 Hz oscillations after a generator trip. A 30 frame/s phasor stream has a 15 Hz Nyquist limit for phasor-time variations, so oscillations near this range can alias or appear as beating.



Harmonics and interharmonics

Standard synchrophasor outputs summarize the fundamental component; harmonic and interharmonic contamination can still dominate estimator failure modes.

Source: Kaua'i 18-20 Hz IBR oscillation analysis; NERC IBR monitoring white paper; IEC/IEEE 60255-118-1

1. Research Problem

Event-aware frequency metrics for low-inertia grids.

Research problem

- Low-inertia IBR grids require reliable local frequency estimates during ramps, phase jumps, harmonic distortion, ringdown, and mixed dynamic events.
- Classical estimators often trade off responsiveness, noise rejection, latency, and computational load.
- Modern model-based and data-driven estimators improve some regimes, but can be difficult to compare fairly.
- Existing comparisons often under-specify seeds, disturbance families, tuning policy, trip-risk, structural latency, or computational cost.

Research gap

The literature needs a reproducible benchmark that connects estimator error to protection-oriented risk, settling behavior, and deployability.

2. Benchmark Gap

Standards, datasets, estimator evidence.

Benchmark literature gap

IEC/IEEE PMU standards → NIST metrology → Open PMU platforms → Real PMU datasets → Algorithm studies

IEC/IEEE PMU standards	NIST metrology	Open PMU platforms	Real PMU datasets	Algorithm studies
Compliance for synchrophasor, frequency, and ROCOF measurements	Traceable calibration and PMU performance assessment	Device and streaming infrastructure for PMU research	Operational events from real grids	Deep studies of selected estimator families

- The field is mature in standards, metrology, data collection, and individual estimator studies.
- The missing layer is a reproducible, estimator-level benchmark that compares many algorithms under identical synthetic truth and deployment metrics.

Source: IEC/IEEE 60255-118-1:2018; NISTIR 8106; OpenPMU; openPDC; FNET/GridEye; LBNL Open micro-PMU; PNNL open PMU event library

Benchmark gaps in prior work

Prior benchmark type	Strength	Gap for estimator-family comparison
IEC/IEEE PMU compliance	Defines synchrophasor, frequency, ROCOF, static/dynamic compliance tests	Leaves algorithm choice, family comparison, and CPU/latency outside the estimator report.
NIST PMU assessment/calibration	Provides metrological evaluation of PMU devices against standard requirements	Focuses on instruments and compliance behavior rather than a large matrix of open estimator implementations under controlled tuning policy.
OpenPMU / openPDC ecosystem	Enables open PMU construction and synchrophasor stream processing	Infrastructure-oriented, with no controlled estimator matrix using matched seeds and stress sweeps.
Open real PMU datasets	Supply realistic events for detection and application development	Truth is partially latent; hard to isolate causal stress factors or compute estimator error against known $f_{\mathrm{true}}[k]$.
Single-study algorithm comparisons	Give detailed evidence for selected methods or scenarios	Usually limited estimator sets, limited artifact contract, and incomplete deployability metrics.

Compliance tests define requirements, operational data define realism, and controlled synthetic truth isolates estimator behavior under reproducible stress.

Benchmark comparison axes

Evidence source	Known truth	IBR composite stress	Trip-risk	CPU/latency	Reproducible artifacts
IEC/IEEE-style compliance	Yes	Limited	No	No	Device-dependent
Field PMU/DFR data	No	Yes	Possible	No	Dataset-dependent
Single-method studies	Often	Partial	Partial	Partial	Often limited
Estimator benchmark	Yes	Yes	Yes	Yes	Yes

Measured outputs

The same voltage trace is scored by RMSE, peak error, RoCoF error, trip exposure, settling, CPU, and structural latency.

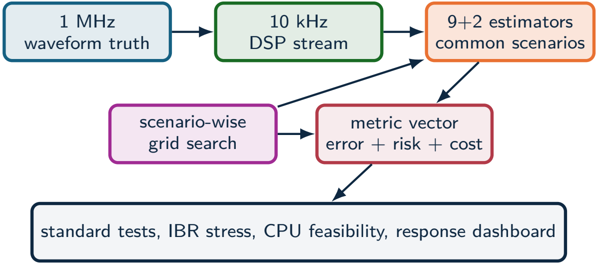
Interpretive effect

An estimator can pass standard cases and still fail a composite IBR event; it can also be accurate but too delayed or costly for protection-adjacent use.

3. Experimental Method

Signals, scenarios, I/O, metrics.

Experimental method



Source: Methodology and test framework

Measurement setting

Local, non-time-synchronized relay-class frequency tracking from a decimated 10 kHz stream; PMU timestamp certification is outside scope.

Experimental principle

Each estimator–scenario pair is tuned within a declared grid, then compared using RMSE, peak error, trip exposure, settling, and CPU.

RA-EKF mechanism

State augmentation

$$x_k = [\theta_k, \omega_k, A_k, \dot{\omega}_k]^\top, \qquad \omega = 2\pi f$$

$$\theta_k = \theta_{k-1} + \omega_{k-1}\Delta t + \frac{1}{2}\dot{\omega}_{k-1}\Delta t^2$$

$$\omega_k = \omega_{k-1} + \dot{\omega}_{k-1}\Delta t$$

IBR-specific stress response

$$Q_k = \mathrm{clip}(r_k, 0.25, 4)\, Q_{\mathrm{base}}, \qquad r_k = \frac{|\nu_k|}{\hat{\sigma}_\nu}$$

- **Covariance scaling**: inflate process noise under transients, deflate near steady state.
- **Event gating**: when $|\nu_k| > \gamma\hat{\sigma}_\nu$, softly reset θ_k while preserving ω_k and $\dot{\omega}_k$.

RA-EKF extends the EKF baseline with an explicit RoCoF state and event-aware covariance behavior for phase-jump recovery and steady-state tracking.

Metrics and decision variables

Error and recovery

$$e[k] = \widehat{f}[k] - f_{\mathrm{true}}[k]$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_k e[k]^2}$$

$$E_{\mathsf{max}} = \max_k |e[k]|$$

Protection-oriented risk

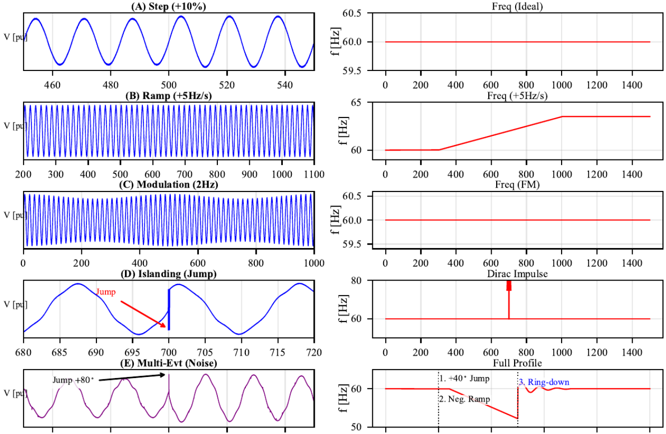
$$T_{\mathrm{trip}} = \Delta t \sum_k \mathbf{1}\{|e[k]| > 0.5\ \mathrm{Hz}\}$$

$$t_s = \min\{t : \ |e[\tau]| \leq \epsilon,\ \forall \tau \geq t\}$$

$$C_{\mathrm{CPU}} = \mathrm{mean\ time/sample}$$

The 0.5 Hz threshold is a protection-oriented risk proxy. The metric set separates average error, transient severity, trip exposure, settling, and compute cost.

Scenario suite



Scenario structure

- A–C are standard-style references.
- D introduces phase discontinuity and harmonic stress.
- E stacks the event mechanisms that standards isolate.
- RMSE, T_{trip}, and CPU need a joint reading.

Estimator families

Loop-based

SRF-PLL, SOGI-FLL.

Window-based

IpDFT, TFT.

Model-based / stochastic

EKF, UKF, RA-EKF.

Adaptive and data-driven

VFF-RLS, Koopman (RK-DPMU), PI-GRU.

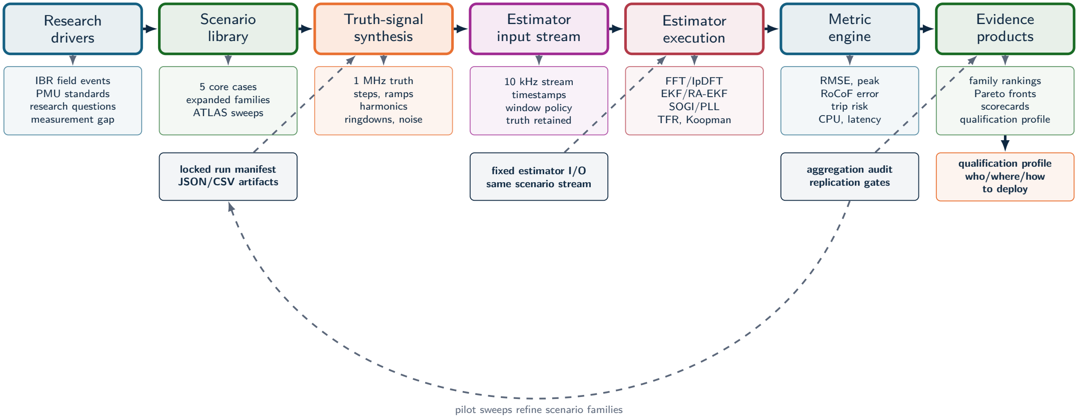
Legacy baselines

RLS and TKEO are retained as comparison references outside the recommended deployment set.

Interpretation rule

These families occupy different latency and stress-sensitivity positions. The comparison tracks where each family wins, degrades, and fails.

Full benchmark workflow



4. Benchmark Evidence

Standard cases, IBR stress, CPU cost, and deployment limits.

Standard scenario results

Method	Step +10% RMSE	Step +10% Peak	Ramp +5 Hz/s RMSE	Ramp +5 Hz/s Peak	Modulation 2 Hz RMSE	Modulation 2 Hz Peak
IpDFT	0.00157	0.00741	0.0729	0.157	0.000281	0.000747
TFT	0.00210	0.0190	0.0726	0.127	0.000260	0.000781
Koopman	**0.000207**	0.00523	0.0549	0.0860	**0.000212**	0.00178
SRF-PLL	0.000261	0.00132	0.142	0.354	0.000238	**0.000441**
SOGI-FLL	0.00900	0.0100	2.12	3.05	0.00900	0.0100
EKF	0.000733	0.00857	0.0180	0.0631	0.00119	0.00265
UKF	0.000774	0.00851	0.0207	0.0657	0.00116	0.00271
RA-EKF	0.00231	0.0299	**0.0113**	**0.0441**	0.00476	0.0125

Key result: RA-EKF is the strongest ramp tracker in this table, with 0.0113 Hz RMSE, 12.6× below SRF-PLL and 1.59× below EKF.

Source: `Table III`

Composite islanding: 275× trip-exposure reduction

Scenario D evidence

- Instantaneous $+60^\circ$ phase jump with harmonic and interharmonic content.
- Windowed leakage: IpDFT reaches 0.481 Hz RMSE; TFT reaches 2.74 Hz.
- EKF/UKF incur $T_{\mathrm{trip}} = 0.165$ s without phase-reset mechanisms.
- SRF-PLL is competitive on RMSE, 0.0645 Hz, with $T_{\mathrm{trip}} = 0$.

RA-EKF result

$$\mathrm{RMSE} = 0.0695\ \mathrm{Hz}$$

$$T_{\mathrm{trip}} = 0.6\ \mathrm{ms}$$

$$0.165\ \mathrm{s}/0.0006\ \mathrm{s} \approx 275\times$$

The gain comes from explicit $\dot{\omega}$ modeling plus event gating rather than post-hoc filtering.

Protection reading: RA-EKF converts long EKF false-trip exposure into a sub-millisecond transient under the islanding phase jump.

Multi-event trip-risk

Method	RMSE [Hz]	T_{trip} [s]	CPU [μs]
RA-EKF	0.511	0.141	54
EKF	0.789	**0.118**	18
Koopman	**0.476**	0.137	85
UKF	0.792	0.119	139
SRF-PLL	0.400	0.471	11
IpDFT	1.36	0.335	48
SOGI-FLL	0.568	0.422	25

Main reading

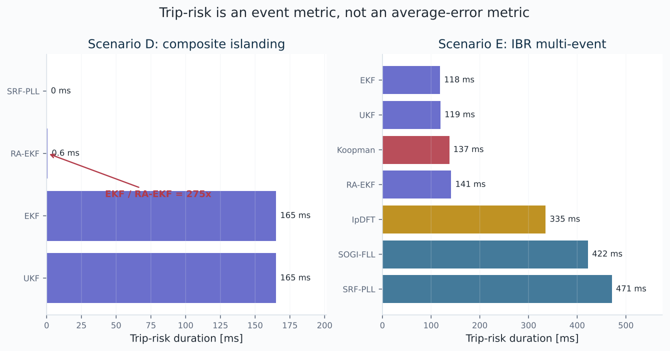
RA-EKF sits in the practical protection region: low trip exposure, moderate RMSE, and embedded-feasible CPU.

Scalar ranking failure

SRF-PLL has low CPU and good RMSE but much higher T_{trip}; EKF has lower T_{trip} but worse RMSE; Koopman is accurate but costlier.

Source: `Table IV`

Trip-risk evidence: core stress cases



Result reading

- Scenario D exposes phase-reset behavior: RA-EKF reduces EKF exposure from 165 ms to 0.6 ms.
- Scenario E is multi-objective: RMSE and trip-risk select different methods.
- SRF-PLL is strong in Scenario D but reaches 471 ms in Scenario E.

Result: T_{trip} separates protection exposure from average error.

Result comparison: winners depend on the question

Question	Best-supported answer	Evidence	Boundary
Clean step/modulation accuracy	Koopman and SRF-PLL are strongest point cases	Step RMSE 0.000207 Hz; modulation peak 0.000441 Hz	Not enough for islanding or multi-event survival
Ramp and RoCoF tracking	RA-EKF is strongest in the standard ramp	Ramp RMSE 0.0113 Hz	Not a universal RMSE winner
Composite islanding exposure	RA-EKF removes long EKF exposure	0.6 ms vs 165 ms for EKF	SRF-PLL is also strong in this specific phase-jump case
Multi-event RMSE	Koopman and SRF-PLL are competitive	0.476 Hz and 0.400 Hz	RMSE hides trip-risk and compute trade-offs
Embedded protection feasibility	SRF-PLL, EKF, SOGI-FLL, RA-EKF remain plausible	11–54 μs/sample range	Python timing is a software-cost proxy

Selection changes with event family, objective metric, and compute envelope. A single leaderboard hides the deployment trade-off.

Computational feasibility

Method	Time/sample	Class	Target
SRF-PLL	11 μs	P2	MCU
EKF	18 μs	P1	DSP
TFT	25 μs	P1	DSP
SOGI-FLL	25 μs	P2	MCU/DSP
IpDFT	48 μs	M1	DSP/FPGA
RA-EKF	**54 μs**	**P1**	**DSP/FPGA**
Koopman	85 μs	M1	CPU
UKF	139 μs	P1	high-perf CPU
PI-GRU	≈136,000 μs	–	GPU only

Source: `Table V`

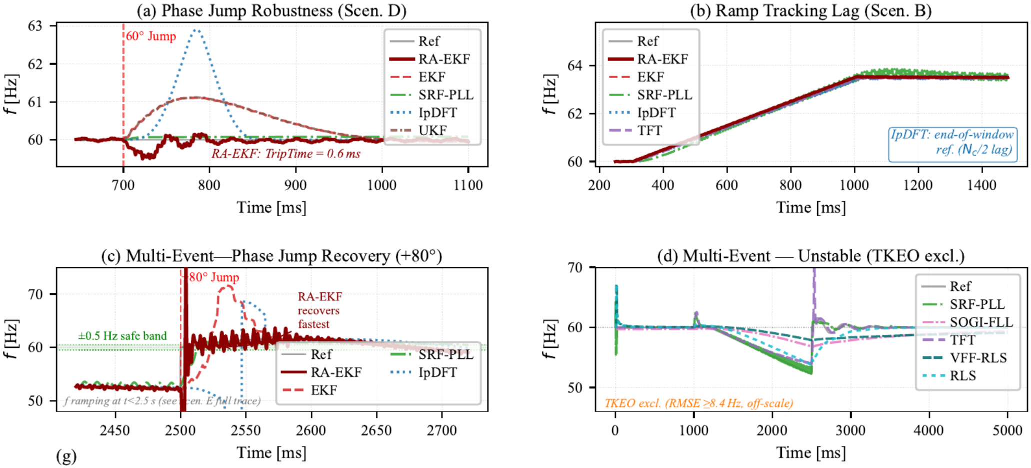
Deployment interpretation

The Python baseline separates feasible estimator families from clearly prohibitive ones; hardware latency requires platform timing.

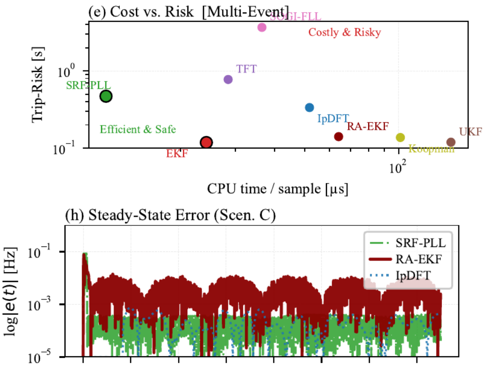
Key comparison

RA-EKF is roughly three times EKF cost in Python but remains below the millisecond scale by a wide margin; PI-GRU is retained as a computational reference outside embedded protection targets.

Event responses



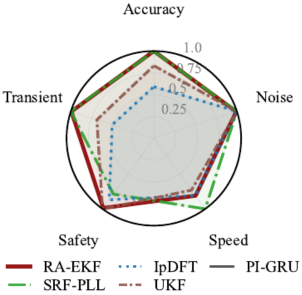
Deployment limits



Dashboard reading

- Deployment risk requires RMSE, trip exposure, settling, and delay.
- Trip-risk exposes the cost of estimator transients in composite events.
- Windowed methods can be accurate but delayed.
- RA-EKF occupies the main protection-oriented trade-off point.

Balance profile



Radar profile

- Accuracy, speed, noise immunity, safety, and transient behavior are conflicting axes.
- RA-EKF is selected for event-rich protection use cases.
- SRF-PLL remains valuable when cost and simplicity dominate.
- IpDFT is attractive where observation delay is acceptable.

Compliance heatmap: why standard tests are insufficient

Method	Step	Ramp	Mod	Isl	Multi
PI-GRU	F	F	F	F	F
Koopman	P	F	P	F	F
TFT	P	F	P	F	F
IpDFT	P	F	P	F	F
SOGI-FLL	P	F	P	P	F
SRF-PLL	P	F	P	F	F
EKF	P	P	P	F	F
UKF	P	P	P	F	F
RA-EKF	P	P	P	F*	F

Several methods pass the standard-style Step/Ramp/Mod columns and still fail under composite islanding or multi-event IBR stress. Standard tests leave IBR event readiness partially untested.

Empirical claims

Protection behavior

RA-EKF turns EKF exposure from long to sub-cycle.
Scenario D gives $T_{\mathrm{trip}} = 0.6$ ms for RA-EKF versus 0.165 s for EKF, while preserving an explicit dynamic state model.

Metric disagreement

RMSE, trip-risk, and CPU select different winners.
Table IV separates Koopman/SRF-PLL accuracy, EKF/UKF trip exposure, and low-cost loop/filter deployment.

Standard-test limitation

Simple-case pass/fail leaves event readiness unresolved.
The heatmap shows methods that pass Step/Ramp/Mod but fail islanding or multi-event IBR stress.

Deployability

Feasibility changes the accuracy ranking.
RA-EKF remains in a 54 μs/sample software-cost regime; PI-GRU is retained as a computational reference outside relay-class timing.

Guardrail: report scenario, metric, and compute envelope before naming a preferred estimator.

Composite IBR sequence: why it is hardest

Scenario E stress components

- Harmonic-rich steady state: 5th at 5%, 7th at 3%, low interharmonics.
- $+40^\circ$ phase jump.
- -6 Hz/s RoCoF segment.
- $+80^\circ$ phase jump with ringdown.
- Sparse impulsive noise.

Standard-test gap

- The estimator sees a nonstationary waveform with stacked disturbances.
- Spectral leakage, loop re-lock, state-model mismatch, and recovery delay occur together.
- A method can be good on RMSE and still undesirable on trip exposure or CPU.

> Once composite stress is shown to matter, ATLAS sweeps each stress component systematically instead of treating the multi-event case as a single aggregate disturbance.

Expanded stress analysis

EXPANDED BENCHMARK: 16 estimators, 34 scenarios, 60 Monte Carlo runs per scenario

Core experiment

- 9 principal estimators + 2 legacy baselines.
- 5 scenarios: A–E.
- Standard, islanding, multi-event, and CPU results.
- RA-EKF supplies the tested estimator mechanism.

Expanded benchmark

- 16-estimator, 34-scenario, 10-family benchmark.
- 60 Monte Carlo runs per scenario in the expanded dataset.
- ATLAS severity curves and harmonic sweeps.
- Used to test sensitivity, scaling, and estimator-family stability.

The expanded benchmark scales the same measurement question across more event families, estimator types, and Monte Carlo replications.

Expanded benchmark: family winners

EXPANDED BENCHMARK: 16 estimators, 34 scenarios, 60 Monte Carlo runs per scenario

Family	Count	RMSE leader	RMSE	Trip-risk leader
IEEE ramps	9	SOGI-PLL	19.4 mHz	0 s, multiple estimators
Magnitude steps	7	EKF	0.388 mHz	0 s, multiple estimators
Modulation	3	RA-EKF	17.8 mHz	0 s, multiple estimators
Phase jumps	3	SOGI-FLL	174 mHz	ZCD, 0.0194 s
OOB interference	1	EKF	13.6 mHz	EKF / RA-EKF, 0 s
IBR harmonics	3	EKF	70.5 mHz	TFT / Prony, 0.00947 s
IBR ringdown	5	UKF	313 mHz	ESPRIT, 0.0989 s
IBR multi-event	1	PLL	173.5 mHz	IpDFT, 0.0439 s

The expanded sweep confirms the central pattern: estimator rankings depend on disturbance family and metric target.

ATLAS severity analysis

ATLAS SEVERITY SWEEP: pilot run, $n = 1$, default controller policy

Core cases → ATLAS severity curves → sign asymmetry → fixed vs oracle policy → validation gate

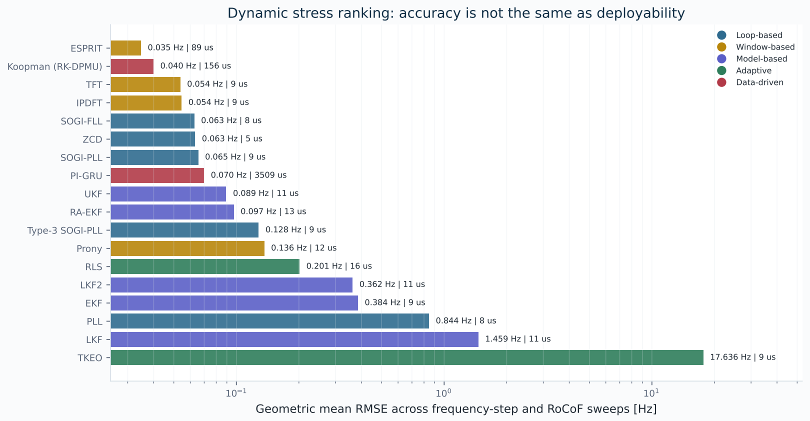
Core cases	ATLAS severity curves	sign asymmetry	fixed vs oracle policy	validation gate
A–E failure mode	When does error start to grow?	Are positive and negative events symmetric?	Does tuning policy change the rank?	Which results replicate?

> ATLAS maps where estimator behavior changes with severity, sign, tuning policy, and replication depth.

Source: `docs/ATLAS_PIPELINE.md`; `docs/ATLAS_METHOD_AUDIT.md`

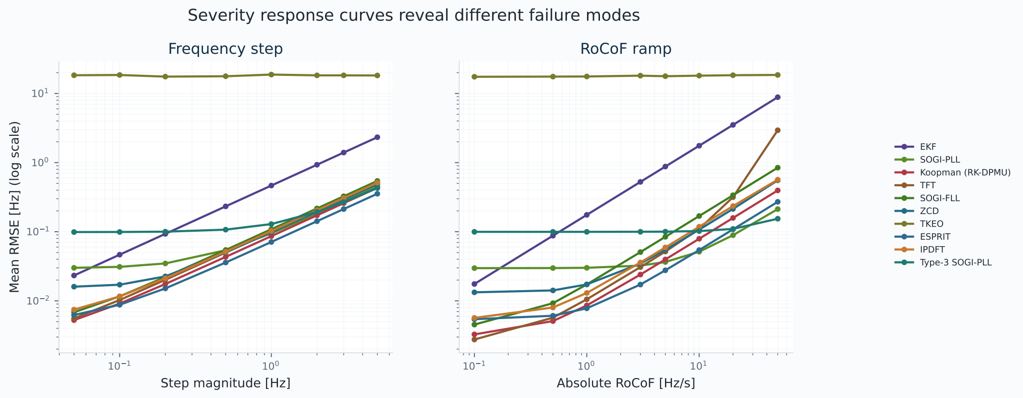
Dynamic accuracy ranking

ATLAS SEVERITY SWEEP: pilot run, $n = 1$, default controller policy



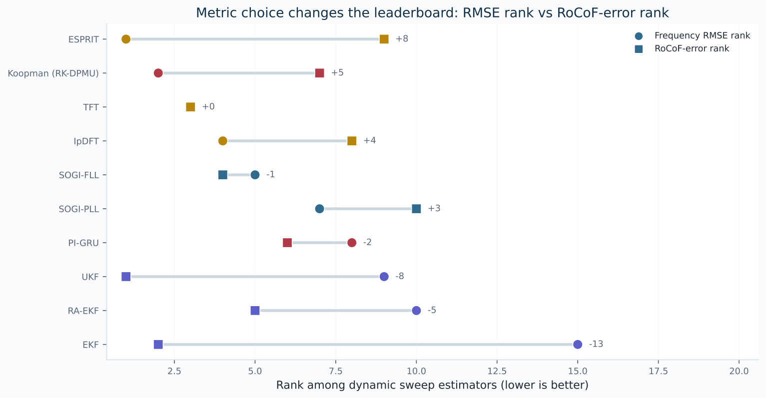
Sensitivity to disturbance severity

ATLAS SEVERITY SWEEP: pilot run, $n = 1$, default controller policy



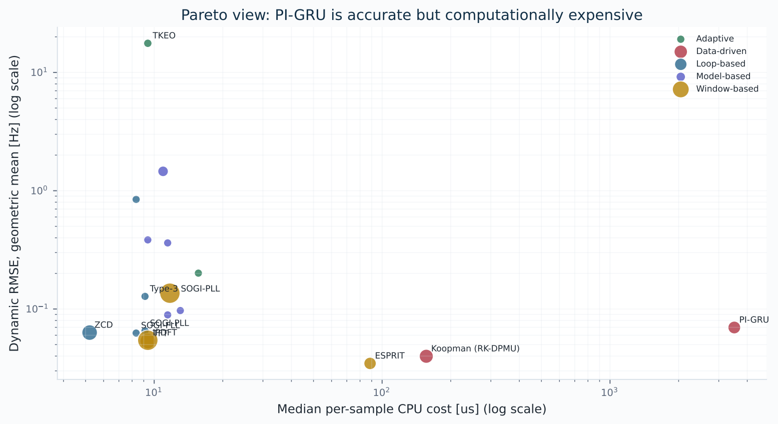
Rank shift when the metric changes

ATLAS SEVERITY SWEEP: pilot run, $n = 1$, default controller policy



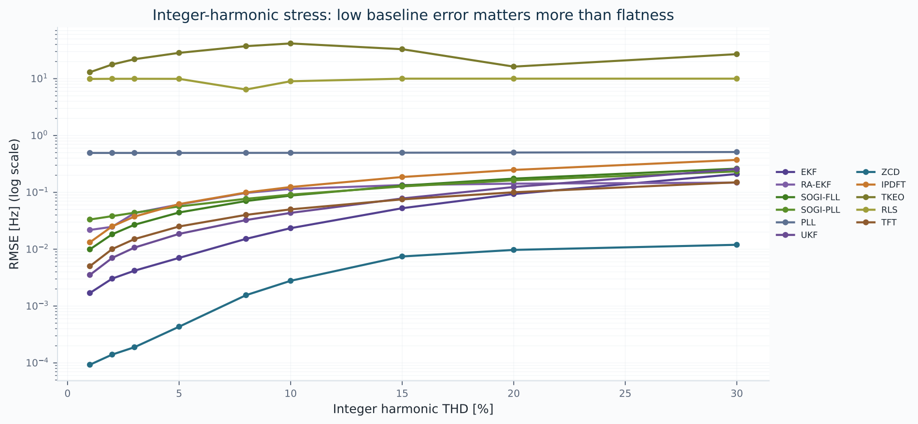
CPU–accuracy Pareto analysis

ATLAS SEVERITY SWEEP: pilot run, $n = 1$, default controller policy



Harmonic-stress response

ATLAS SEVERITY SWEEP: pilot run, $n = 1$, default controller policy



5. Insights and Validity

Rankings, recommendations, validity.

Three findings

1. **Latency and stress trade-off**: IpDFT/TFT provide strong steady-state accuracy but carry 3–5 cycle structural delay; loop methods are fast but can suffer elevated trip exposure.
2. **State-space estimation matters**: RA-EKF's explicit $\dot{\omega}$ state plus event gating reduces Scenario-D trip exposure from 0.165 s for EKF to about 0.6 ms.
3. **Compliance tests are insufficient**: estimators that pass standard-style Step/Ramp/Mod cases can fail under composite islanding and multi-event IBR stress.

Estimator qualification includes transient recovery, trip exposure, compute feasibility, and standard-case RMSE.

Validity limits and next steps

Validation limits

- Limited to the reported scenario set.
- CPU is a software-level cost indicator; hardware latency requires platform timing.
- The study is simulation-only.
- Scenario-wise tuning can affect rank stability.

Validation program

- Hardware-in-the-loop validation.
- Wider converter-control interaction coverage.
- Sensitivity analysis of the tuning protocol.
- ATLAS severity and sign sweeps with validation gates.

These limits define the validation path: broader event coverage, hardware timing, fixed-policy replication, and field replay.

Recommendation map

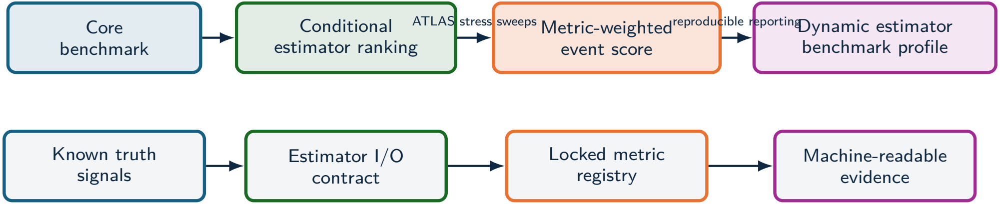
- FFR / RoCoF tracking → RA-EKF explicit $\dot{\omega}$
- Anti-islanding phase jumps → RA-EKF event gating
- Low-cost relay core → SRF-PLL / EKF CPU efficient
- Metering-like steady state → IpDFT / TFT if delay acceptable

Deployability depends on event class, trip-risk tolerance, and compute budget.

6. Deployment Profile

Qualification logic and next validation.

Qualification profile



Estimator selection moves toward **qualification** when each method is tested against a declared event profile, metric profile, and latency-cost envelope.

Standardization profile

Current standardization layer

- PMU standards define phasor, frequency, and RoCoF measurement requirements.
- Compliance tests are necessary, with estimator-family ranking by IBR event survival left outside scope.
- Device reports often hide algorithmic failure modes behind aggregate accuracy.

Standardization boundary

A dynamic-estimator profile needs declared signals, estimator I/O, metrics, evidence artifacts, and latency-cost reporting.

Benchmark profile

- Canonical events: steps, ramps, phase jumps, ringdown, harmonics, interharmonics, noise, and composite IBR sequences.
- Locked metrics: RMSE, peak error, RoCoF error, trip-risk, settling, invalid samples, CPU, and latency.
- Task profiles: forensic monitoring, PMU augmentation, DFR support, and relay-adjacent screening.

Standardization value

A reference test profile needs the same signals, estimator I/O, metrics, and machine-readable evidence.

IBR Event Qualification Score

COMPOSITE EVENT SCORE: weighted event-class qualification metric

Score for estimator e

$$\mathrm{IBR\text{-}ERS}(e) = 1 - \sum_{s \in \mathcal{S}} w_s \, \Phi_s(e),$$

$$\Phi_s(e) = \alpha \tilde{E}_s + \beta \tilde{P}_s + \gamma \tilde{T}_s + \delta \tilde{S}_s + \eta \tilde{C}_s + \lambda \tilde{L}_s .$$

- $\tilde{E}_s$: normalized dynamic RMSE.
- $\tilde{P}_s$: normalized peak deviation.
- $\tilde{T}_s$: trip-risk duration.
- $\tilde{S}_s$: settling and recovery penalty.
- $\tilde{C}_s, \tilde{L}_s$: compute and latency penalties.

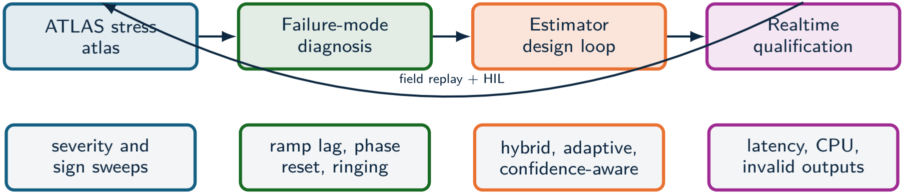
Composite metric

RMSE, trip-risk, transient recovery, and cost lead to different decisions. A composite score states the deployment objective explicitly.

Profile-specific weights

Protection-adjacent use can raise γ, δ, λ; forensic reconstruction can raise α, β; embedded monitoring can raise η.

Toward realtime IBR estimators



- The stress analysis identifies which physical stressor breaks each estimator family.
- That diagnosis supports hybrid estimators: event classifiers, harmonic-aware preprocessing, adaptive bandwidth, and self-reported confidence.
- Validation should combine synthetic truth, EMT replay, hardware-in-the-loop, and field oscillography.

Validation agenda

Near-term extensions

- Full ATLAS sweeps with fixed policy and 100-run Monte Carlo replication.
- Statistical rank stability under tuning, noise, and disturbance severity.
- Composite qualification profiles for forensic, PMU-augmentation, and relay-adjacent tasks.

Longer-term research program

- Three-phase, unbalance, and converter-control interaction profiles.
- Real-event replay from DFR/PMU/oscillography with synchronized ground-truth proxies.
- Estimators that optimize an objective vector: frequency error, RoCoF, trip exposure, settling, CPU, and latency.

Next validation step: evaluate each estimator against declared event regimes, latency limits, and compute envelopes.

Conclusion

- Field disturbances define the measurement problem; controlled synthetic truth makes estimator behavior explainable.
- The study compares estimators across objectives, regimes, and deployment constraints, including IBR-relevant stress modes.
- Empirical conclusions remain tied to evidence level, stress regime, and replication depth.

Result

Estimator choice is tied to the event class, metric target, and compute envelope before deployment.